



Heavy liquid metal natural circulation in a one-dimensional loop

G. Coccoluto^{a,1}, P. Gaggini^{a,1}, V. Labanti^{a,1}, M. Tarantino^{a,*,1}, W. Ambrosini^{b,2},
N. Forgiione^{b,2}, A. Napoli^{b,2}, F. Oriolo^{b,2}

^a ENEA Brasimone, 40032 Camugnano (BO), Italy

^b Università degli Studi di Pisa, Dipartimento di Ingegneria Meccanica, Nucleare e della Produzione Via Diotisalvi, 2 - 56126 Pisa, Italy

ARTICLE INFO

Article history:

Received 5 March 2010

Received in revised form 18 June 2010

Accepted 25 June 2010

ABSTRACT

The ENEA Brasimone Research Centre since 1999 is strongly involved in the national and European research programmes performed in the field of heavy liquid metal technology aiming at the development of critical (LFR) and subcritical (ADS) nuclear systems.

In particular, in the frame of the IP-EUROTRANS, (6th Framework Program EU), ENEA assumed the commitment to perform an integral experiment with the aim to reproduce the primary flow path of a HLM pool-type nuclear reactor, cooled by lead bismuth eutectics (LBE). This new experimental activity, named ICE “Integral Circulation Experiment”, will be performed by an appropriate test section designed to be installed in the CIRCE facility.

In order to support the ICE activity, as well as characterize the natural and gas enhanced circulation flow regimes in a HLM loop, qualify test procedures and components which are nuclear relevant, a new facility was designed and built-up by Brasimone Research Centre, named NACIE “NATural Circulation Experiment”.

The paper reports a detailed description of the loop and the main experimental results carried out from the natural circulation tests already performed on the NACIE loop.

Numerical simulations have been performed in collaboration with the University of Pisa, adopting the RELAP5/Mod3.3 system code modified to allow for LBE as a cooling fluid.

The aim of the performed post-test calculations is to compare the code response with the experimental results under the natural circulation flow regime, allowing to qualify the adopted nodalization as well as the performance of the code when employed on HLM loop.

© 2010 Elsevier B.V. All rights reserved.

1. Introduction

In the last years, wide importance is given to the activities in support of the development of systems for the transmutation of radioactive waste, as can be noticed from the European Union (EU) projects EUROTRANS (EUROTRANS, 2005) “EUROpean research programme for the TRANsmutation of high level nuclear waste in Accelerator Driven Systems”, which concurs to demonstrate the feasibility of an ADS-type dedicated transmuter. Other activities are ongoing with the aim to develop a Generation IV LFR concept (GIF, 2002), like ELSY (ELSY, 2006) “European Lead-Cooled System” project, aimed at showing the possibility of realization and operation of a safe and competitive fast lead-cooled critical reactor.

In parallel, project dedicated to address specific issues have been and still are carried out, such as, for example, considering the

EU framework programmes, TECLA “TECHnologies for Lead Alloys”, dedicated to lead technologies, aimed at assessing the use of lead alloys both as a spallation target and as a coolant for an ADS, VELLA (VELLA, 2006) “Virtual European Lead Laboratory”, aimed at integrating the existing European HLM infrastructures, developing synergies and complementarities among the laboratories and the research groups across the EU.

In this framework, several R&D priorities, and planned efforts, have been defined, focusing on system design, fuels development, HLM technology, materials, component development, demonstration (Cinotti et al., 2004, 2006; Fazio et al., 2005).

In particular large-scale integral tests to characterize the behaviour of the main systems are necessary especially for supporting the licensing process.

An integral system experiment is foreseen in ENEA, where the circulation of a sector of the primary lead coolant in the reactor pool will be reproduced. The normal steady state conditions operating transients, (e.g. pumping system start-up and core power changes) and incidental transients, (e.g. transient from forced to natural circulation) will be performed in the experimental facility.

* Corresponding author. Tel.: +39 0534 801262; fax: +39 0534 801250.

E-mail address: mariano.tarantino@enea.it (M. Tarantino).

¹ Tel.: +39 0534 801262; fax: +39 0534 801250.

² Tel.: +39 050 836657; fax: +39 050 836665.

Nomenclature

Roman letters

C_p	specific heat [J/(kg K)]
D	diameter [m]
f	friction factor
g	gravity [m/s ²]
q''	heat flux [W/m ²]
Q	heating power [W]
T	temperature [°C]
w	fluid velocity [m/s]
W	mass flow rate [kg/s]

Subscripts

HS	heat source
LBE	lead bismuth eutectic

Abbreviations

ADS	accelerator driven system
EC	European commission
EU	European union
FPS	fuel pin simulator
HLM	heavy liquid metal
HS	heat source
HX	heat exchanger
HTC	heat transfer coefficient
ICE	integral circulation experiment
LBE	lead bismuth eutectic
LFR	lead fast reactor

Acronyms

CIRCE	Circolazione Eutettica
EUROTRANS	EUROpean research programme for the TRANSmutation of high level nuclear waste in accelerator driven systems
ELSY	European Lead-Cooled System
NACIE	NATural Circulation Experiment
TECLA	TEchnologies for Lead Alloys
VELLA	Virtual European Lead Laboratory

This experimental activity, named “Integral Circulation Test”, ICE, will be performed by the ENEA C.R. Brasimone, where an appropriate test section has been designed to be installed in the CIRCE facility (Tarantino and Scaddozzo, 2006; Tarantino, 2008).

CIRCE is a multipurpose facility located by the Brasimone Research Centre, built-up with the aim to perform thermal-hydraulic experiments with heavy liquid metal coolant, support the definition and to allow the qualification of chemistry control system for pool system, to test prototypical component for the HLM technology, to run large-scale experiment in a pool configuration (Turroni et al., 2001).

CIRCE basically consists of a main cylindrical vessel (S100) with an outer diameter of 1200 mm and an height of 8500 mm. The main vessel can be filled with about 70 tons of molten lead bismuth eutectic (LBE), and it has been designed to host different test sections welded to and hung from bolted vessel heads for the study of the thermal-hydraulic issues related to the HLM pool systems. The main vessel is moreover equipped with auxiliary systems for eutectic circulation, with argon cover gas and recirculation system, LBE heating and cooling systems.

In Fig. 1 an isometric view of the CIRCE facility is shown, as well as the ICE test section.

The ICE activity will allow getting detailed information about different topics, as to evaluate the thermal-hydraulics behaviour

of a HLM pool system by the analysis of the coupling between an appropriate heat source and a cold sink placed inside it, to investigate the behaviour of representative components, (i.e. prototypical heat exchangers) for the LFR/ADS concepts, to characterize the natural circulation flow regime in a HLM pool system.

In this frame, the NACIE loop facility has been built-up to carry out experimental tests needed to characterize the natural and gas enhanced circulation flow regimes, and to qualify components for HLM applications, especially in support of the ICE activities.

In fact, since the heat source designed for the ICE activity, named fuel pin simulator FPS, is based on the use of prototypical electrical heaters, a preliminary characterization of their thermal and electrical performance is required; this characterization will be accomplished by the NACIE loop.

1.1. NACIE loop description

As already mentioned, the aim of NACIE loop is to set-up a support facility able to qualify and characterize components, systems and procedures relevant for HLM nuclear technologies.

Moreover, by this facility it will be possible to perform several experimental campaigns in the field of the thermal-hydraulics, fluid-dynamics, chemistry control, corrosion protection and heat transfer, allowing to obtain correlations essential for the design of the nuclear power plant cooled by heavy liquid metal.

Finally, the possibility to test four prototypical fuel pin simulators as well as all their ancillary systems and mechanical connections is mandatory in order to confirm the design of the ICE test section.

For this reason the NACIE loop is prepared to house a bundle made with four prototypical fuel pin simulators in full scale to the ones which will be manufactured for the ICE test section.

NACIE is a heavy liquid metal rectangular loop which basically consists of two vertical pipes (I.D. 62.7 mm) working as riser and downcomer, connected by means of two horizontal branches (I.D. 62.7 mm).

The adopted material is stainless steel (AISI 304) and the total inventory of LBE is about 1000 kg; the design temperature and pressure are 550 °C and 10 bar respectively.

In the bottom part of the riser a heat source is installed through an appropriate flange, while the upper part of the downcomer is connected to a heat exchanger.

The difference in level between the thermal centre of the heat source and the one of the heat sink was fixed to reproduce the same height that characterizes the ICE test section ($H = 4.5$ m).

The loop is completed by an expansion vessel, installed on the top part of the loop, coaxially to the riser.

The general layout of the loop is shown in Fig. 2, while the geometrical data, which characterize NACIE facility, are given in Table 1.

For what concern the instrumentation, 13 thermocouples type-k (accuracy 2%) are installed along the main stream line to monitoring the liquid metal temperature in different relevant point of the loop. Other thermocouples (four, type-k) are installed along the water

Table 1
NACIE main geometrical data.

Total vertical length [mm]	9231
Horizontal length [mm]	1000
Pipe inner diameter [mm]	62.7
Pipe thickness [mm]	5.16
Expansion vessel height [mm]	765
Expansion vessel inner diameter [mm]	254.5
Heat exchanger length [mm]	1500
Heat source active length [mm]	1000

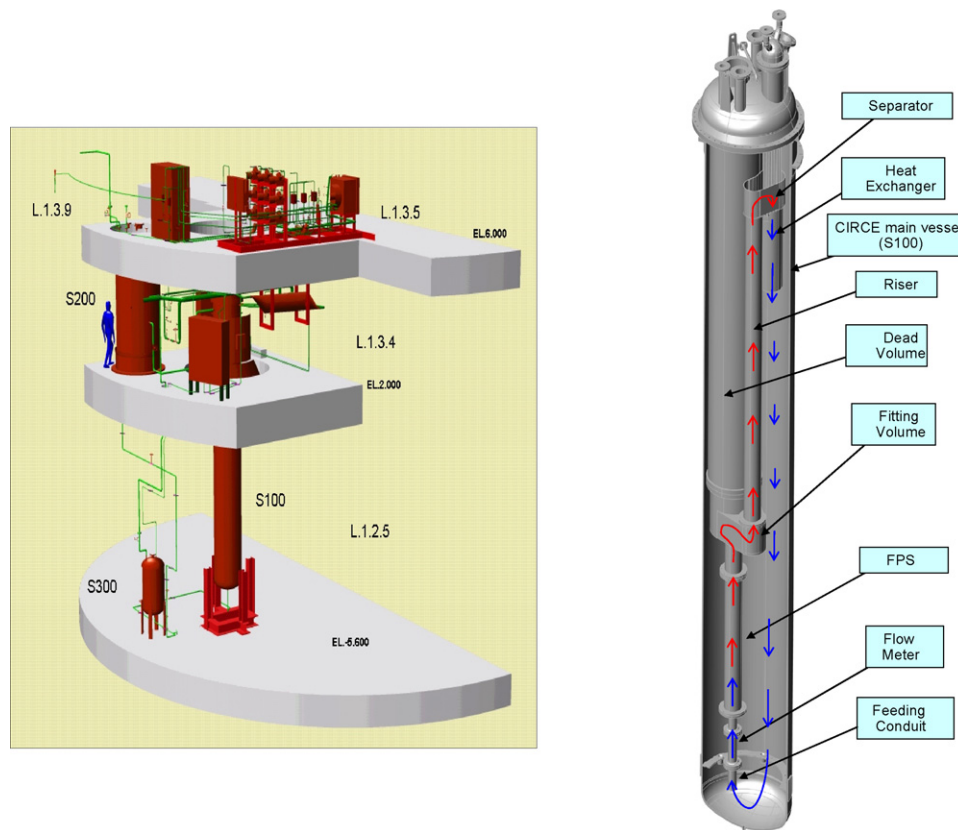


Fig. 1. CIRCE isometric view (left) and ICE test section (right).

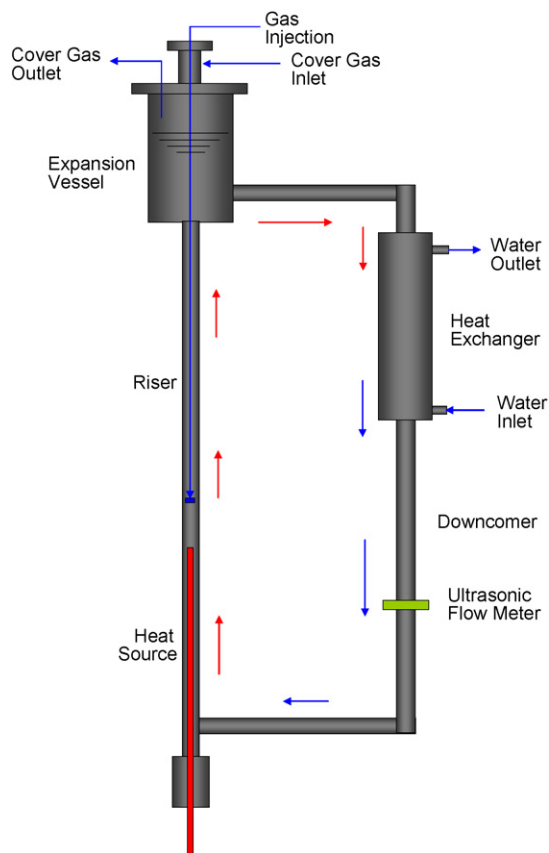


Fig. 2. NACIE loop layout.

circuit as well as into the gas recirculation system (one into the cover gas).

Moreover, an electromagnetic flow meter (accuracy 1%) is placed along the secondary loop to monitor the water flow rate into the heat exchanger, and two level probes (type on-off) are placed into the expansion tank to monitoring the level variation during the filling and draining of the loop.

Finally, along the gas recirculation circuit, which allows to kept constant the argon pressure into the cover gas and the gas injection into the riser for circulation promotion (gas enhanced circulation) three pressure transducers (type strain gauge, accuracy <0.1%) are installed.

The pressure transducers are adopted to monitor the pressure of the argon along the supply line (0–10 bar), into the cover gas (0–500 mbar), and into the gas lift injection pipe (0–8 bar). On the same pipe a gas flow meter (thermal mass flow meter, accuracy <0.2%) is installed to have a direct measurement of the argon injected during the gas enhanced circulation test.

For the NACIE activity two different bundles are foreseen, named “high flux bundle” and “low flux bundle”.

The first experimental campaign will be run adopting the high flux bundle, with the goal to characterize prototypical heating elements, supporting the implementation of the large-scale experimental named ICE (Tarantino and Scaddozzo, 2006; Tarantino, 2008).

For the manufacturing of the ICE large fuel pin bundle (thermal power of 1 MW) two different technological solution have been proposed, named A-type and B-type respectively (see Table 2).

The first solution (coaxial structure), allow to get a uniform distribution of the heat flux around the fuel pin, while the second one (bifilar structure) a not-uniform distribution.

The high heat flux bundle after the run of the first tests will be removed, installing the so-called low flux bundle. The objective is

Table 2
High flux bundle characteristics.

A-type prototypical fuel pin	
Technology	Coaxial
Number	2
Diameter [mm]	8.2
Active length [mm]	1000
Total length [mm]	2760
Heat flux [W/cm ²]	100
Flux distribution	Uniform
Thermal power [kW]	25
B-type prototypical fuel pin	
Technology	Bifilar
Number	2
Diameter [mm]	8.2
Active length [mm]	1000
Total length [mm]	8380
Heat flux [W/cm ²]	100
Flux distribution	Non-uniform
Thermal power [kW]	25

to carry out natural and gas enhanced circulation tests aiming to investigate the phenomena related with the mixed convection in rod bundle cooled by heavy liquid metal.

1.2. High flux bundle

The high flux bundle consists of four prototypical fuel pins. A view of the bundle is shown in Fig. 3 while its main characteristics are given in Table 2.

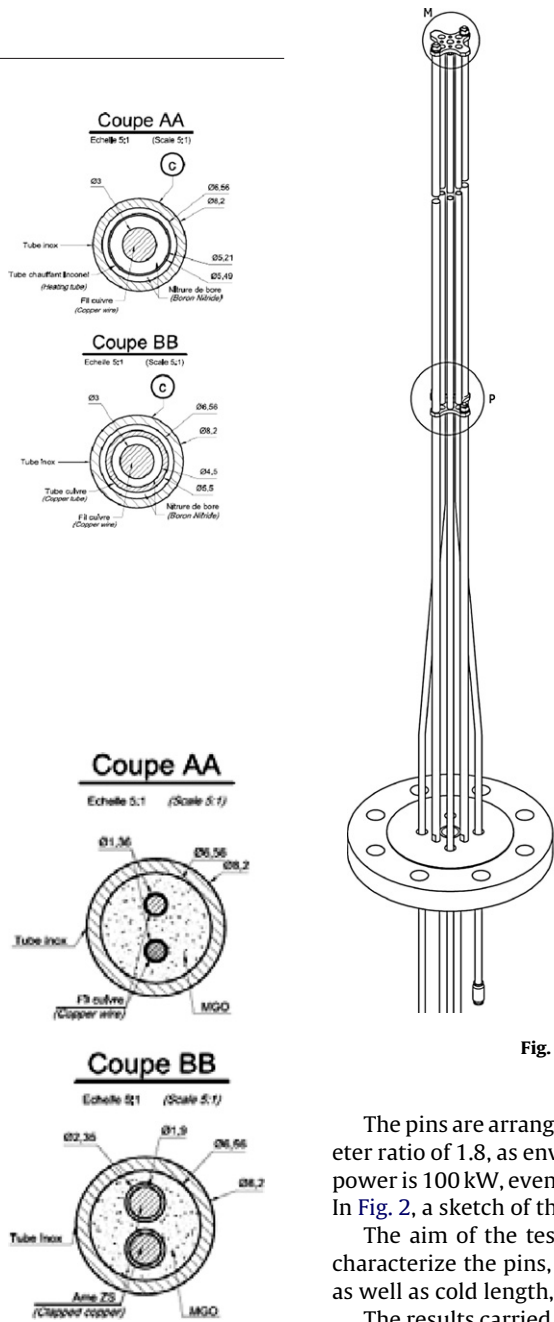


Fig. 3. View of the high flux bundle.

The pins are arranged in a triangular lattice with a pitch to diameter ratio of 1.8, as envisaged for the ICE bundle. The total installed power is 100 kW, even if during the tests only one pin will be active. In Fig. 2, a sketch of the high flux bundle is reported.

The aim of the tests planned using the high flux bundle is to characterize the pins, in term of performance of the active length as well as cold length, reliability and thermal fatigue.

The results carried out should allow choosing the more suitable technical solution for the pins to adopt in the ICE bundle.

In particular, the results carried out for the B-type prototypical fuel pins have been reported into the paper, being the first option proposed for the ICE activity.

In fact, B-type technological solution allows to built-up pins with a cold length of several meters (see Table 1), simplifying the handling of the high current electrical connectors in a pool system.

1.3. Low flux bundle

The low flux bundle consists of three electrical cartridges placed on triangular lattice with a pitch to diameter ratio of 1.4; the total thermal power installed is 30 kW.

The main parameters of the low flux bundle are reported in Table 3.

The low flux bundle will be instrumented to allow HTC measurement under natural circulation flow regime as well as under gas-injection enhanced circulation.

Table 3
Low flux bundle characteristics.

Number	3
Diameter [mm]	19.05
Active length [mm]	1000
Total length [mm]	2380
Heat flux [W/cm ²]	16.8
Flux distribution	Uniform
Thermal power [kW]	10

The instrumentation proposed for the HTC measurement consist of several thermocouples placed along the main subchannel (in the centre of the low flux bundle). In particular three measurement section are foreseen, at the inlet, middle and outlet section of the heating length.

For each section 3 thermocouples have to be installed on each pin, immersed in the clad (0.5 mm in depth) with a diameter of 0.8 mm. Moreover a thermocouple with a diameter of 1–1.5 mm have to be placed in the centre of the subchannel, along the main streamline.

The possibility to install a Pitot tube in the outlet section of the bundle to make an accurate measurement of the flow rate along the main subchannel will be evaluated. This system never has been proven before on the NACIE loop.

1.4. Pumping system

To promote the LBE circulation along the loop, a gas lift technique will be adopted (Benamati et al., 2007; Ambrosini et al., 2003, 2005; Tarantino, 2007).

A pipe having inner diameter of 10 mm is housed inside the riser, connected through the top flange of the expansion tank to the argon feeding system.

At the other end of the pipe the injection nozzle is mounted, and is located just downstream the heating section. The gas will be injected in the riser through the nozzle, enhancing the liquid metal circulation. In the expansion tank the argon is separated from the gas–liquid mixture flowing upward into the cover gas, while the liquid metal flows along the upper horizontal branch.

In this way the possibility to have a two phase mixture flowing through the heat source is avoided.

The gas injection system is able to supply argon flow rate in the range [1–75 NI/min] with a maximum injection pressure of 5.5 bar.

1.5. Heat exchanger

The heat exchanger (HX) designed for the NACIE loop is a “tube in tube”, counter flow type; the secondary fluid is water at low pressure (about 1.5 bar). The HX is made by three coaxial tubes with different thicknesses; the dimensions of tubes are reported in Table 4.

LBE flows downward into the internal pipe, while water flows upwards into the annulus between the middle and external pipes (see Fig. 4).

The annulus created by the internal and middle pipes, that has a width of 5.95 mm, is filled by stainless steel powder. The aim of this powder gap is to guarantee the thermal flux towards water, since

Table 4
NACIE HX pipes dimension.

	Inner pipe	Middle pipe	External pipe
Inner diameter [mm]	62.68	84.9	102.3
Outer diameter [mm]	73	88.9	114.3
Thickness [mm]	5.16	2.0	6.02
Length [mm]	1500	1500	1500
Material	AISI 304	AISI 304	AISI 304

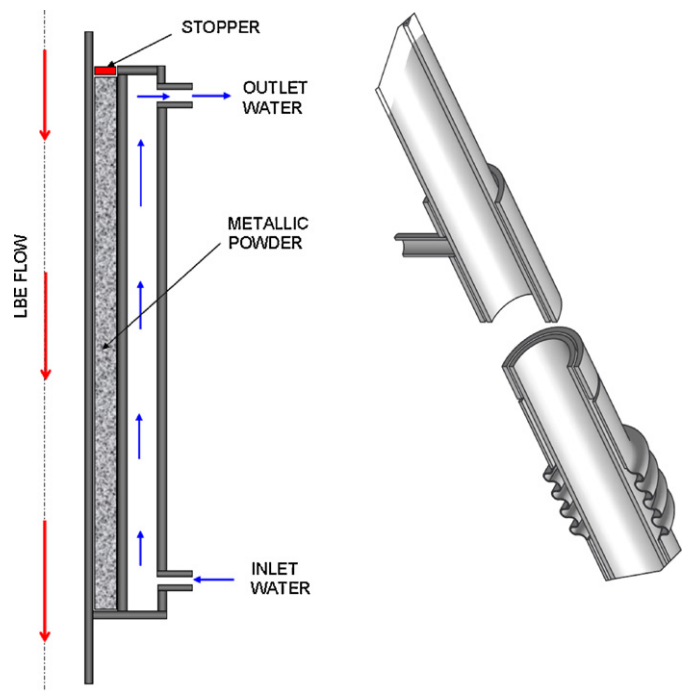


Fig. 4. Scheme of the HX and view of the expansion joint.

it has a good thermal conductivity, mitigating at the same time the thermal stresses on the pipes due to the differential thermal expansion along the axis during the operation. In fact, the three pipes are welded together in the lower part by a plate, while in the upper part only the middle and external pipes are constrained together. In this way the internal pipe has no axial constraints with the HX and axial expansion is allowed.

The annulus containing the powder is closed at the top by a “stopper” made by graphite, in order to avoid the powder leakage.

Finally, on the external pipe an axial expansion joint is installed to compensate the different axial expansion between the middle and external pipe (see Fig. 4).

Moreover, by the use of a powder gap, which mostly contributes to the overall thermal resistance of the component (about 50%), the thermal gradient through the thickness of the pipes is strongly reduced, in spite of the large thermal difference between the water and LBE stream flow.

The NACIE HX has been designed assuming a thermal duty of 30 kW; it is the case of the low flux bundle under the natural circulation flow regime.

2. Relap nodalization for the NACIE loop

The scheme of the nodalization adopted for RELAP5 is reported in Fig. 5. It consists of several pipe, branch and single junction components aiming at the simulation of the different parts of the loop.

The heater power, generated by the prototypical fuel pins simulator, has been imposed in heating rods. Heat losses have been also introduced in the analyses and have been adjusted according to the information coming from the experiments. For this aim, appropriate heat transfer coefficient and fluid temperature are imposed outside the pipe walls, also taking into account the pipe thermal insulation.

Nodes, each of which having 10 cm of pipe length, are introduced throughout the nodalization, wherever possible.

The time dependent volume 60 and the time dependent junction 55 are considered to allow the simulation of the tests with the assisted circulation through the injection of the argon in the riser.

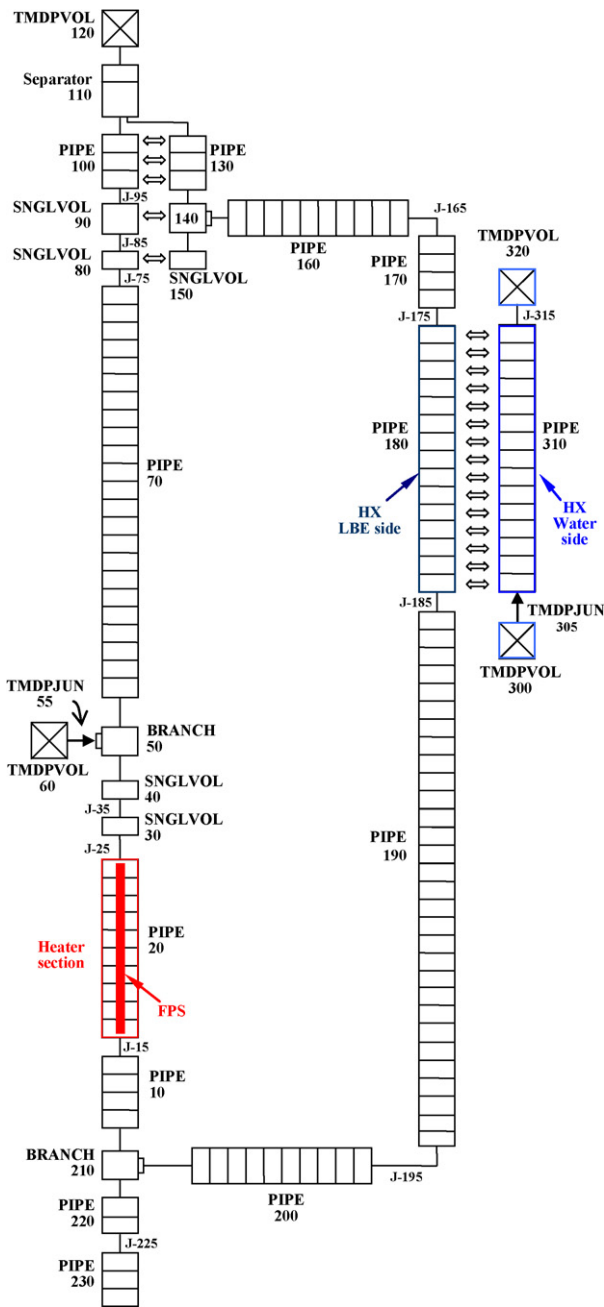


Fig. 5. RELAP5 nodalization for the NACIE loop.

In this work, the injected argon flow rate is maintained equal to zero.

The time dependent junction 305 is used to fix the right water flow rate. The time dependent volume 300 impose the temperature of the water at the inlet section of the HX secondary side, while the time dependent volume 320 impose the outlet pressure.

In order to simulate the effect of the heating cables, which are present on the external pipe surface along the loop and used to avoid the solidification of the LBE when the heater power is off, a high HTC between pipes and the external environment has been imposed.

In order to better reproduce the thermal behaviour of the heat exchanger, especially for what concern the conductivity through the thickness of the powder gap, a proper thermal conductivity has to be introduced in the code, because it deviates widely from the theoretical value as consequence of the powder compactness

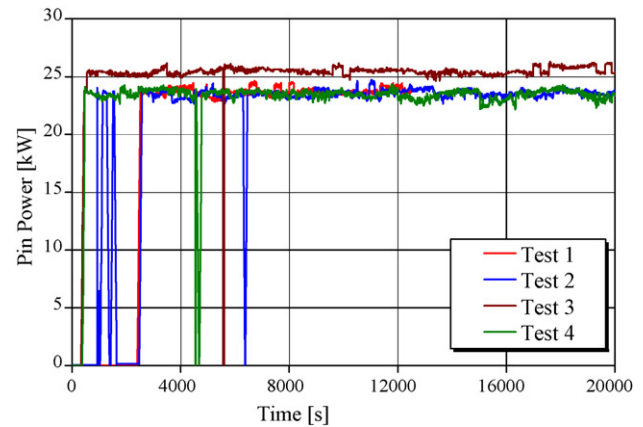


Fig. 6. Electrical power supplied to the loop, by the fuel pin simulator type-B.

achieved during the installation. The powder gap thermal conductivity was evaluated as 12.5% of the stainless steel theoretical value, thanks to the first experimental run performed on NACIE loop, devoted to the HX measurement performance.

3. Performed tests in the NACIE loop

The experimental activity performed with NACIE facility included four different tests concerning natural circulation. In particular, the tests have been performed with only one pin activated, type B, inside the heating section, with a nominal power of 25 kW.

The differences in the heater power for the four tests can be observed in Fig. 6. As it can be noted, the electrical power effectively supplied to the system is about 23.5 kW for tests nos. 1, 2 and 4, nearly 95% of the nominal power, while in the test no. 4 the power is very near the nominal value.

In fact, one of the main activities performed during the run of the first tests were to tune the power supply and control system as well as to set-up the data acquisition system.

As can be noted in Fig. 6, several times the power was shut-down during the test (as for example test 2) allowing to act on the power supply and control system.

Different boundary conditions are instead applied for the water temperature at the inlet section of the HX secondary side and mass flow rate as depicted in Figs. 7–9. In particular, in tests nos. 1 and 2 the inlet temperature of the water has been started at around 65 °C and the decreased to about 25 °C when an average temperature of LBE of about 350 °C is achieved along the heat source (see Fig. 8).

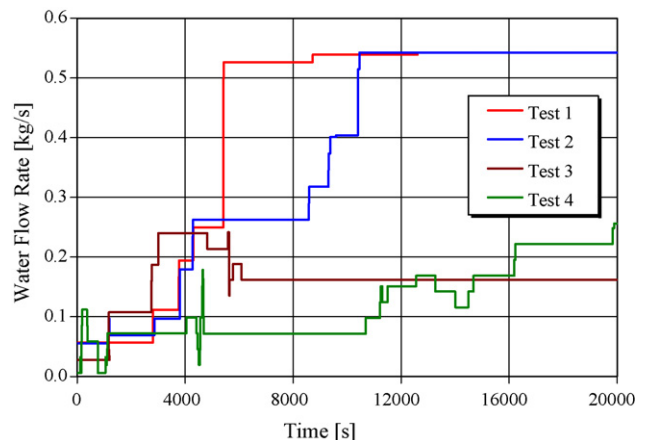


Fig. 7. Time evolution of the water flow rate in the HX secondary side.

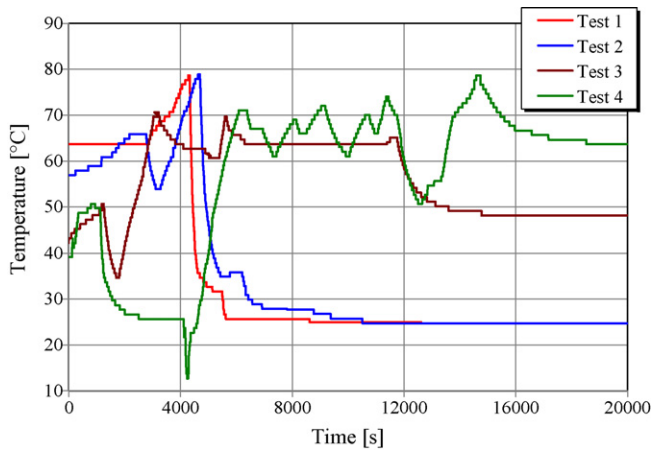


Fig. 8. Measured water temperature in the inlet HX section.

This behaviour is due to the start of air cooler installed in the secondary loop. In the case of tests nos. 3 and 4 the inlet temperature of the water is maintained at about 80 °C trough out the testing period (see Fig. 8).

As for the case of the power supplied to the system, the water flow rate into the heat exchanger, as well as its temperature, was tuned several times during the run of the first natural circulation tests, because, being the facility at the begin of life, no direct data were available about the flow rate needed to promote the natural circulation at well defined average temperature of the LBE.

Fig. 7, and then Fig. 8, underlines that the water flow rate (regulating the performance of the centrifugal pump), as well as its inlet temperature (regulating the performance of the air cooler installed on the water loop) were changed several times aiming to match as practicable as possible the natural circulation test specifications.

Finally, Fig. 7 depicts as during the transient, the secondary pump is active, to avoid the boiling of water inside the heat exchanger, even if the flow rate is set at a value much lower than the nominal value of 0.55 kg/s.

For a direct measurement the LBE flow rate achieved during the natural circulation tests, an ultrasonic flow meter has been chosen as proper instrument, because its good accuracy and reliability. Moreover, because it is a not-intrusive device, the pressure drop along the main flow path will be not increased, optimizing the natural circulation performance of the loop.

For the installation of the ultrasonic flow meter, high temperature wave guide are under development by ENEA, but their performance, in terms of reliability, never have been tested before.

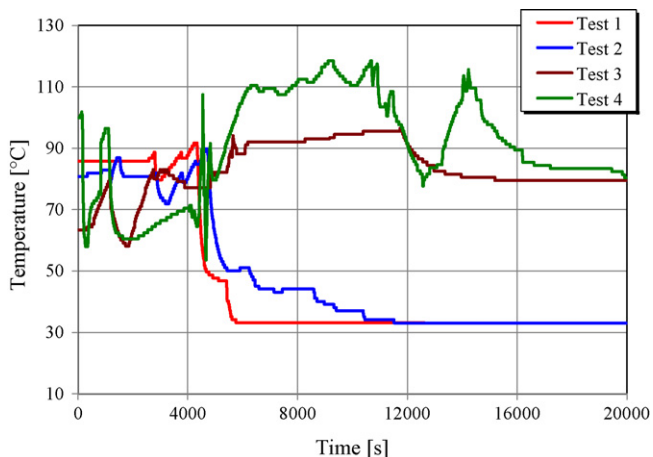


Fig. 9. Measured water temperature in the outlet HX section.

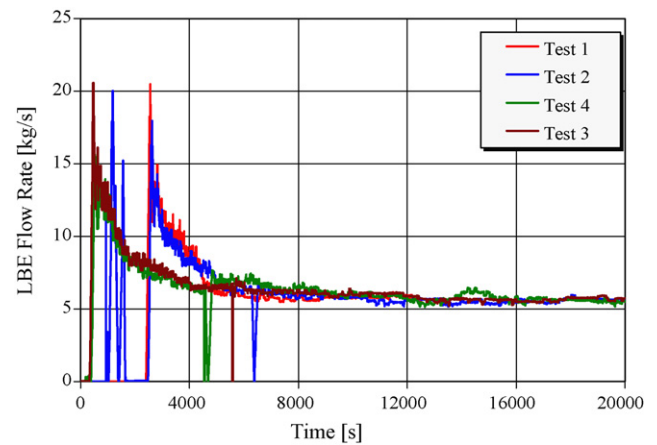


Fig. 10. LBE flow rate calculated by an energy balance through the HS.

For this reason the ultrasonic flow meter is not yet available along the NACIE loop, and so an estimation of the LBE flow rate under steady state condition has been calculated applying the energy balance equation across the heat source, as in the following:

$$W = \frac{\dot{Q}_{HS}}{c_{p,LBE} \Delta T_{HS}} \quad (1)$$

As it can be noted in Fig. 10, the estimated LBE flow rate in the steady state conditions is about 5.5 kg/s, allowing to achieve a liquid metal velocity along the active pin of about 0.18 m/s and a Reynolds number of 6.7E+4.

Moreover Fig. 10 shown as a steady state condition is obtained after about 6000 s, but no information about the flow rate during the transients are available due to the method adopted for the flow rate estimation. In fact, as already mentioned, the energy balance approach does not allow to estimate the flow rate during the transient, because an important role is played by the thermal inertia of the structures.

4. Results

The fluid-dynamic behaviour of the system has been simulated by a version of the RELAP5/Mod3.3 code (RELAP5/Mod. 3 Code Manual, 1999), modified in order to take into account the LBE fluid properties, constitutive equation and correlation for heat transfer evaluation through rod bundle.

In Figs. 11 and 12 are reported the time trends of the calculated and energy balance flow rate, respectively for the test nos.

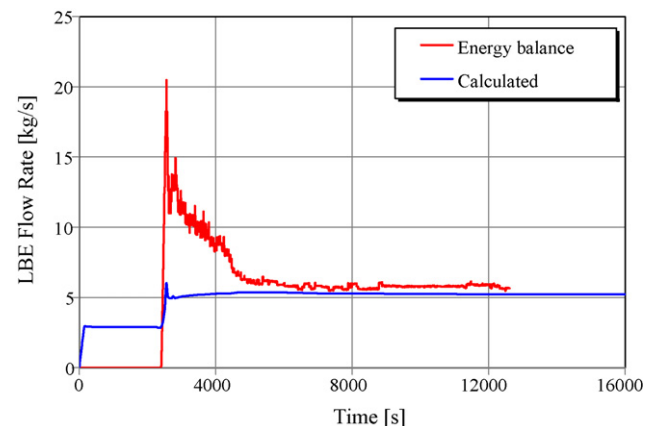


Fig. 11. Comparison between calculated and "energy balance" LBE flow rate for the test no. 1.

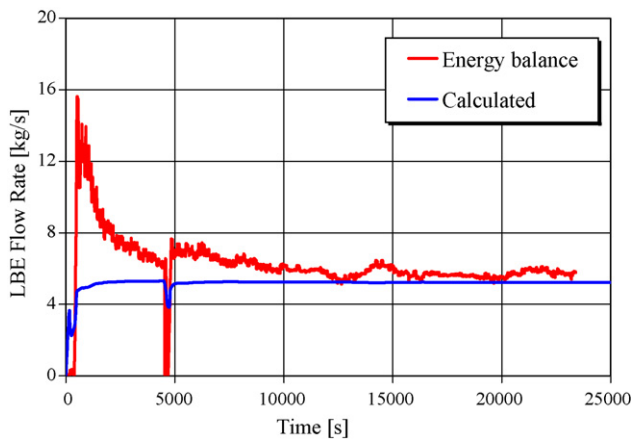


Fig. 12. Comparison between calculated and “energy balance” LBE flow rate for the test no. 4.

1 and 4. There is a very good agreement between calculated and experimental trends, mainly in the last part of the transient, with relationships closely matching with each other. The figures show that in the first phase of the transient the flow rate evaluated by the energy balance through the heat source is considerably higher than that obtained from the calculations. This behaviour is due to the fact that the energy balance is based on the assumption of steady state conditions.

In the first part of the tests, during the transients, the numerical simulations allow to have a good estimation about the expected liquid metal flow rate. As can be noted, the LBE flow rate quickly reach its steady state value, underlines that the fluid-dynamic inertia of the system is really lower than the thermal inertia.

Moreover, in the calculation results can be observed that the LBE flow rate, when the heater power is off (first phase of the transient), is not equal to zero. This is because, before the start of the heating power in the HS the temperature along the loop is maintained constant with the heating cables. In the same time water circulate in the secondary side of the heat exchanger causing a not negligible heat exchange between primary and secondary loops.

The presence of LBE circulation in the loop during the first phase of the transient is confirmed by the experimental trend of the inlet and outlet HX temperatures, reported in Figs. 13 and 14. In fact, both calculations and measures show a temperature difference of the LBE between inlet and outlet HX section also in the first phase of the analysed transient, when the electrical power applied to the heater section is zero, though we will aspect that the flow rate is slightly overestimated by the code.

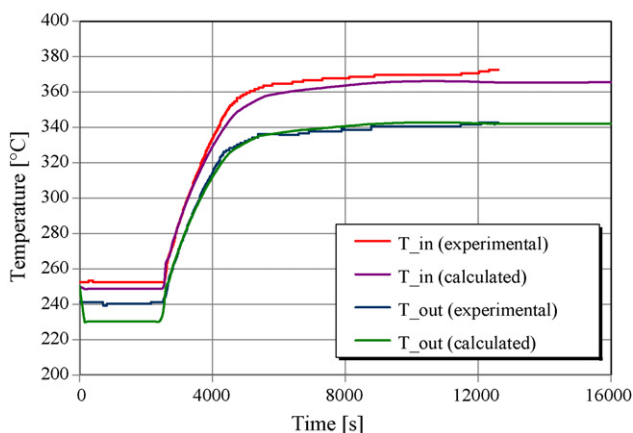


Fig. 13. Comparison between calculated and measured temperatures at the inlet and outlet HX sections for the test no. 1.

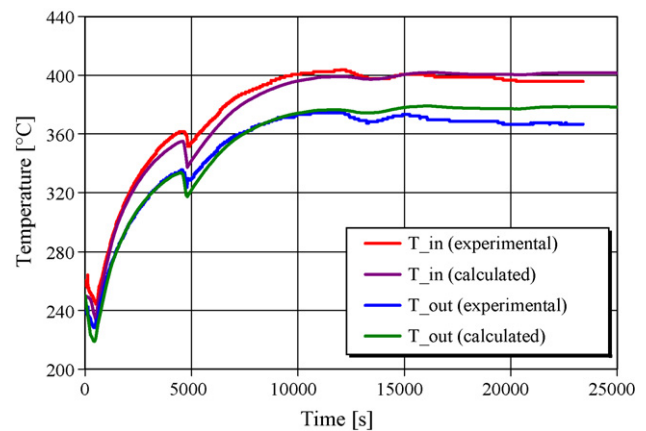


Fig. 14. Comparison between calculated and measured temperatures at the inlet and outlet HX sections for the test no. 4.

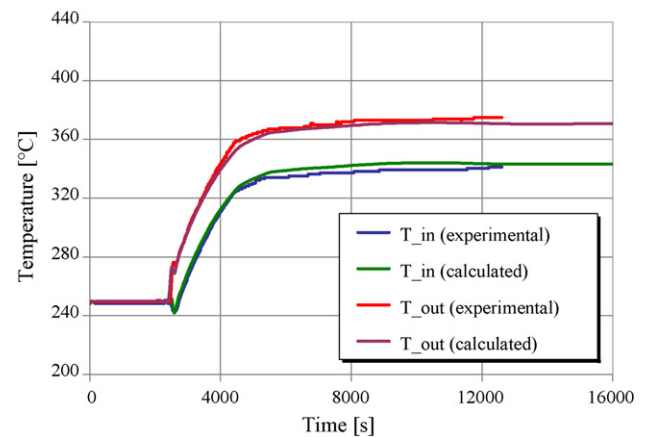


Fig. 15. Comparison between calculated and measured temperatures at the inlet and outlet HS sections for the test no. 1.

Finally, as depicted in Figs. 11–16, the numerical simulations well match the thermal-hydraulic behaviour of the NACIE loop, both on the liquid metal side than on the water side.

These agreements highlight as the parameters adopted to reproduce the powder gap thermal conductivity, the heat transfer coefficient between the structures and the environment, the thermal inertia of the structure, the pressure drops along the main flow path of the LBE, allow to properly fit the behaviour of the NACIE loop.

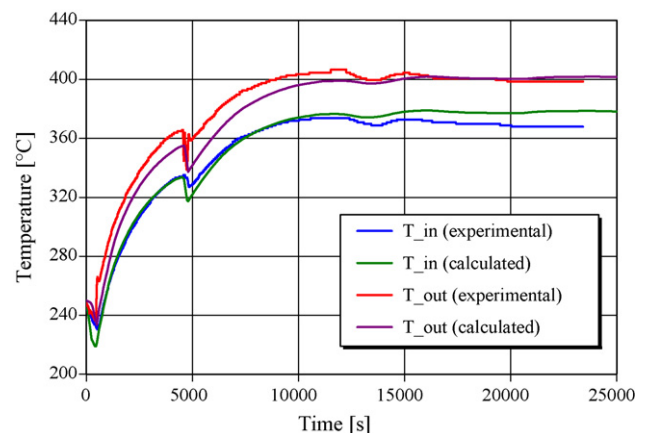


Fig. 16. Comparison between calculated and measured temperatures at the inlet and outlet HS sections for the test no. 4.

Moreover, the performed simulations shown as a good prediction of the system behaviour can be achieved if the right boundary condition is defined especially for what concern the secondary side of the HX.

On these conditions, the RELAP code is a reliable numerical tools which provides a support design for the safety evaluation of innovative nuclear system with feasible efficiency and good accuracy.

5. Conclusions

The NACIE heavy liquid metal loop was designed and built-up by the Brasimone Research Centre to support the ICE test section design.

In fact, due to the high thermal performance required for the ICE Heat Source, prototypical fuel pin simulator has been realized adopting two different technological solution, named type-A (coaxial solution) and type-B (bifilar solution).

The aim of the NACIE loop is to house a high flux bundle, made by four prototypical fuel pins simulator and to test and qualify the elements in order to select which will be the adopted as fuel pin simulator for the ICE bundle.

Experimental tests have been performed by the NACIE loop, testing the data acquisition system, the power supply system, allowing to perform a first thermal characterization of the prototypical fuel pins simulator. In particular, feeding only one fuel pin simulator, steady state natural circulation tests have been run with a thermal power of 25 kW.

The first results show that the NACIE loop behaviour agrees with the design specifications and, also, the thermal behaviour of the prototypical fuel pin simulator matches with the numerical simulations performed by the supplier.

After the high flux bundle qualification tests, a low flux bundle will be installed on the NACIE loop and a new experimental campaign will start. Its aim is to characterize the natural circulation and gas enhanced circulation flow regimes in a HLM loop. Several experimental data on natural circulation heat transfer coefficient in a rod bundle assembly will be available, covering the knowledge gap existing in this field. Finally, several operational and accidental transients will be simulated, allowing to establish a reference experiment for benchmarking of commercial codes (RELAP, CATHARE, etc.) when employed in HLM loop.

Concerning the computer code simulations, RELAP5 Mod3.3 code provide good results for the LBE flow rate and LBE temper-

atures inside the loop. Taking correctly into account inertia of the thermal structures, the RELAP5 Mod3.3 code is also able to reproduce in a good way the transient of the flow rate in the circuit. As soon as the flow meter will be installed in NACIE loop, better comparison will be possible particularly in the initial transient phase of the tests.

The performed activity encourages further developments. In particular, assisted circulation test, with argon injection, will be conducted and experimental data analysed with the RELAP5 Mod3.3 code by using the same nodalization, which is provided in this paper.

References

- Ambrosini, W., Forasassi, G., Forgione, N., Oriolo, F., Tarantino, M., 2003. Natural and gas-injection enhanced circulation in a loop with variable friction. In: International Conference on Global Environment and Advanced Nuclear Power Plants, GENES4/ANP2003, September 15–19. Kyoto Research Park, Kyoto, Japan.
- Ambrosini, W., Forasassi, G., Forgione, N., Oriolo, F., Scaddozzo, G., Tarantino, M., 2005. Experimental study on combined natural and gas-injection enhanced circulation. *Nuclear Engineering and Design* 235 (10–12), 1179–1188.
- Benamati, G., Foletti, C., Forgione, N., Oriolo, F., Scaddozzo, G., Tarantino, M., 2007. Experimental study on gas-injection enhanced circulation performed with the CIRCE facility. *Nuclear Engineering and Design* 237 (7), 768–777.
- Cinotti, L., et al., 2004. The eXperimental Accelerator Driven System (XADS) Designs in the EURATOM. 5th Framework Program. *Journal of Nuclear Materials* 335, 148–155.
- Cinotti, L., Fazio, C., Knebel, J., Monti, S., Ait Abderrahim, H., Smith, C., Suh, K., 2006. Lead-Cooled Fast Reactor, May 16, 2006, FISA 2006, Kirchberg, Luxembourg.
- EC Project ELSY, 2006. Contract number FP-036439.
- EC Integrated Project EUROTRANS, 2005. Contract number FI6W-CT-2004-516520.
- EC Project VELLA, 2006. Contract number FP6-036469.
- Fazio, C., Alamo, A., Almazouzi, A., Gomez-Briceno, D., Groeschel, F., Roelofs, F., Turroni, P., Knebel, J.U., 2005. Assessment of reference structural materials, heavy liquid metal technology and thermal-hydraulics for European waste transmutation ADS. In: Proceedings of GLOBAL, Tsukuba, Japan, Oct 9–13.
- GIF, 2002. "A Technology Roadmap for Generation IV Nuclear Energy Systems", issues by the U.S. DOE Nuclear Energy Research advisory Committee and the Generation IV International Forum, Washington, DC, United States.
- RELAP5/Mod3 Code Manual, 1999. Code Structure, System Models and Solution Methods, vol. I. The Thermal Hydraulics Group, SCIENTECH Inc., Idaho.
- Tarantino M., 2007. Report ENEA HS-F-R-001 Gas Enhanced Circulation Experiments On Heavy Liquid Metal System.
- Tarantino, M., Scaddozzo, G., 2006. Report ENEA ET-F-S-001 Test Specifications of the Integral Circulation Experiment (ICE).
- Tarantino M., 2008. Report ENEA ET-F-S-002 Integral Circulation Experiment. Intermediate Report.
- Turroni, P., Cinotti, L., Corsini, G., Mansani, L., 2001. The CIRCE Facility", AccApp'01&ADTTA'01, Nuclear Application in the New Millennium, Reno (NV, USA). November 11–15, 2001.